

Quantum Computing and Quantum Machine Learning

Summary FYS5419/9419

Department of Physics, University of Oslo

Spring semester 2026

Course Philosophy and Structure

Two-Part Programme

- 1 **Quantum Computing** (Jan–Mar)
Many-body quantum systems, VQE, fault-tolerant algorithms
- 2 **Quantum Machine Learning** (Apr–May)
NISQ-era variational methods, QNNs, QPINNs, Boltzmann machines

Central Algorithm

The **Variational Quantum Eigensolver (VQE)** links both halves of the course.

Key Themes

- Hilbert spaces & quantum states
- Entanglement & measurement
- Variational circuits (NISQ)
- Fault-tolerant algorithms
- Quantum-enhanced ML
- Solving ODEs/PDEs with quantum methods

Week 1 (Jan 19–23) Basic Notions of Quantum Mechanics

Mathematical Framework

- Linear algebra reminder
- Hilbert spaces and operators
- Bra-ket notation: $|\psi\rangle$, $\langle\phi|$
- Inner products & norms

Quantum States

- One-qubit systems
- Pure states vs. mixed states
- Composite systems
- Operator algebra on Hilbert spaces

Foundation

This week establishes the mathematical language used throughout the entire course.

Weeks 2–3 (Jan 26 – Feb 6) Composite Systems, Density Matrices & Measurements

Week 2: Tensor Products & Entanglement

- Spectral decomposition
- Measurement formalism
- Density matrices $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$
- Entanglement: pure & mixed states

Week 3: Density Matrices in Depth

- Reduced density matrices
- Partial traces
- Von Neumann entropy
- Entanglement measures

Key Concept: Entanglement

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad \Rightarrow \quad S(\rho_A) = \log 2$$

Week 4 (Feb 9–13) Entanglement, Entropies & Quantum Gates

Entanglement & Entropies

- Review of density matrices
- Entanglement entropies
- Separability criteria

Quantum Gates (Introduction)

- Single-qubit gates: X , Y , Z , H , S , T
- Two-qubit gates: CNOT, CZ, SWAP
- Gate universality
- Simple Hamiltonian systems

Bridge to VQE

Understanding gates and Hamiltonians prepares for **Project 1**: first exposure to the VQE algorithm.

Week 5 (Feb 16–20) Getting Started with the VQE Algorithm

Quantum Gates & Simple Algorithms

- Quantum circuit model
- Parameterised circuits (ansätze)
- Gate decompositions
- One- & two-qubit Hamiltonians

VQE Overview

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle \geq E_0$$

Minimise $E(\theta)$ over circuit parameters θ .

Project 1 Launch

Implement VQE for simple one- and two-qubit Hamiltonians.

Key tasks:

- Design ansatz circuit
- Evaluate energy expectation
- Classical optimisation loop

Week 6 (Feb 23–27) Implementing VQE: Measurements & Gradients

Advanced VQE Topics

- **Parameter-shift rule** for gradients:

$$\partial_{\theta} E = \frac{E(\theta + \frac{\pi}{2}) - E(\theta - \frac{\pi}{2})}{2}$$

- Adaptive VQE (ADAPT-VQE)
- Shot-noise and measurement strategies
- Simulation codes and debugging

Hamiltonians Explored

- One-qubit spin models
- Two-qubit interacting systems
- Mapping physics to qubits
- Introduction to the **Lipkin model**

Weeks 7–8 (Mar 2–13) VQE for the Lipkin Model & Jordan-Wigner

Week 7: Lipkin Model VQE

- Two-qubit Hamiltonian implementation
- Full VQE pipeline for the Lipkin collective model
- Benchmarking against exact diagonalisation
- Circuit optimisation strategies

Week 8: Fermionic Hamiltonians

- Lipkin model: final results
- **Jordan-Wigner transformation:**

$$a_j \rightarrow Z^{\otimes j} \otimes \sigma^-$$

- Mapping fermionic operators to qubits
- Beyond the Lipkin model

Milestone

Project 1 results discussed; students have a working VQE solver.

From NISQ to Fault-Tolerant Algorithms

- Project 1 final discussions
- Discrete Fourier Transform review
- **Quantum Fourier Transform (QFT):**

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle$$

- Circuit decomposition: $O(n^2)$ gates vs. $O(n 2^n)$

QFT Circuit Structure

- Hadamard gates
- Controlled phase rotations R_k
- Bit-reversal permutation
- Exponential speedup over classical FFT

Weeks 10–11 (Mar 23 – Apr 10) QFT, Quantum Phase Estimation & HHL

Quantum Phase Estimation (QPE)

- QFT as subroutine
- Estimating eigenvalues of U :

$$U|u\rangle = e^{2\pi i\varphi}|u\rangle$$

- Circuit construction & precision
- Applications to chemistry & physics

HHL Algorithm (Linear Systems)

- Solving $Ax = b$ on a quantum computer
- Uses QPE as subroutine
- Exponential speedup (sparse, well-conditioned A)
- Implementation and codes
- Limitations and caveats

Note

No classes March 30 – April 3 (public holiday in Norway).

Week 12 (Apr 13–17) HHL Algorithm: Codes & Implementation

Deep Dive into HHL

- Full circuit implementation
- Ancilla qubit rotation: eigenvalue inversion
- Amplitude encoding of b
- Uncomputation and measurement
- Error analysis and success probability

HHL Complexity

$$\mathcal{O}\left(\log N \cdot \kappa^2 \cdot \frac{1}{\epsilon}\right)$$

vs. classical $\mathcal{O}(N^3)$

where κ = condition number,
 ϵ = precision,

N = matrix dimension.

Transition to Quantum Machine Learning

With QFT, QPE and HHL covered, we now turn to the **NISQ-era QML** methods for the remainder of the course.

Week 13 (Apr 20–24) QAOA: Quantum Approximate Optimisation

QAOA Algorithm

- Combinatorial optimisation problems
- Alternating cost and mixer unitaries:

$$|\gamma, \beta\rangle = U_M(\beta_p) U_C(\gamma_p) \cdots U_M(\beta_1) U_C(\gamma_1) |+\rangle$$

- Classical outer-loop optimisation
- Depth- p approximation ratio

Implementation Focus

- Mapping problems to Ising Hamiltonians
- Circuit construction
- Gradient evaluation
- Benchmarking on toy instances

Connection

QAOA is the optimisation counterpart to VQE and serves as a gateway into Quantum Machine Learning methods.

Week 14 (Apr 27 – May 1) Quantum Machine Learning: Foundations

QML Basics

- What is QML? Taxonomy of approaches
- Data encoding strategies: amplitude, angle, IQP, etc.
- Parameterised quantum circuits as models
- Expressibility and entangling capacity
- Barren plateaus and trainability

Quantum Neural Networks

- QNN architecture overview
- Layered variational circuits
- Loss functions & gradients
- Links to classical deep learning
- Connections to QAOA

Weeks 15–16 (May 4–15) QPINNs: Solving Differential Equations

Physics-Informed QNNs

- Classical PINNs review
- **Quantum PINNs (QPINNs):** replace neural network with a parameterised quantum circuit
- Encoding PDEs as loss functions
- Automatic differentiation on circuits

Applications & Implementation

- Solving ODEs and PDEs: Schrödinger, heat, Poisson
- Boundary condition encoding
- Gradient computation via parameter-shift rule
- Codes, benchmarks, and Week 16 deep-dive into implementations

Key Idea

QPINNs marry the physics-informed training paradigm with the expressive power of quantum circuits to solve differential equations on near-term hardware

Week 17 (May 18–22) Quantum Boltzmann Machines

Quantum Boltzmann Machines (QBM)

- Classical RBM / BM recap
- Quantum generalisation: thermal states as model distributions
- Training via quantum log-likelihood gradients
- Transverse-field Hamiltonians as model parameters
- Restricted QBM (RQBM) architecture

Connections & Outlook

- Links to VQE and variational circuits
- Generative quantum models
- Applications in physics & optimisation
- Open research problems

Course Conclusion

QBMs illustrate how quantum statistical mechanics and machine learning merge—a fitting capstone for the course

Course Road Map at a Glance

Weeks	Dates	Topics
1	Jan 19–23	Basic quantum mechanics, Hilbert spaces, qubits
2–3	Jan 26–Feb 6	Composite systems, density matrices, entanglement
4	Feb 9–13	Entanglement entropies, quantum gates
5–6	Feb 16–27	VQE: algorithm, measurements, gradients (Project 1)
7–8	Mar 2–13	VQE for Lipkin model, Jordan-Wigner transform
9	Mar 16–20	Quantum Fourier Transform
10–11	Mar 23–Apr 10	QPE, HHL algorithm (<i>holiday Apr 1–3</i>)
12	Apr 13–17	HHL codes and implementation

Summary: What You Will Learn

Part 1 – Quantum Computing

- Formalism of quantum mechanics
- Entanglement & density matrices
- Variational algorithms (VQE)
- Fault-tolerant algorithms:
QFT, QPE, HHL
- Physical models: Lipkin,
fermionic Hamiltonians

Part 2 – Quantum ML

- QAOA for optimisation
- Quantum Neural Networks
- Physics-Informed QNNs
(QPINNs)
- Quantum Boltzmann Machines
- Connections to classical ML

Overarching Goal

Equip physics graduate students with the theoretical foundations and practical implementation skills to design, analyse, and run quantum algorithms on near-term and fault-tolerant hardware